

Quantitative Finance

for Absolute Beginners

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“Written as I wish someone had taught me.”

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Chapter 6

Probability — The Language of Uncertainty

“Probability is the very guide of life.”

— Cicero

Scenario: Will the Market Go Up Tomorrow?

A junior analyst at a hedge fund walks into the morning meeting and announces: “I’m 100% sure the market will go up today.”

The portfolio manager raises an eyebrow. “Nobody is 100% sure of anything in markets. What’s your *probability estimate*?”

“Well... maybe 65%.”

“Good. Now we can work with that. If you’re 65% confident, your position size should reflect that uncertainty.”

This is the essence of quantitative finance: **every decision is made under uncertainty, and probability is the language we use to quantify that uncertainty**. Whether you’re pricing an option, measuring risk, or building a trading strategy, probability theory is the foundation.

6.1 What Is Probability?

Probability = Quantified Uncertainty

Probability assigns a number between 0 and 1 to every possible outcome:

- $P = 0$: Impossible (the sun won’t rise tomorrow)
- $P = 1$: Certain (you will eventually die)
- $P = 0.5$: Equally likely as not (fair coin lands heads)

Key insight: Probabilities must sum to 1 over all possible outcomes. If there are only two outcomes (up or down), and $P(\text{up}) = 0.65$, then $P(\text{down}) = 0.35$.

6.1.1 The Three Axioms of Probability

All of probability theory rests on just three axioms, stated by Kolmogorov in 1933:

 Interview: Estimation

Q: Why is it hard to estimate expected returns?

A:

The **standard error of the mean return** is $\sigma / \sqrt{n}$. With annual volatility $\sigma = 20\%$ and $n = 10$ years:

$$SE = \frac{20\%}{\sqrt{10}} = 6.3\%$$

A 95% CI for the mean return is roughly $\pm 12.6\%$ —enormous relative to a typical equity premium of 5–8%.

Why volatility is easier: Volatility can be estimated from **daily data**. With 252 days/year, 10 years gives $n = 2,520$ observations. The SE of the volatility estimate is roughly $\sigma / \sqrt{2n}$, which is tiny.

Implication: In portfolio optimization, uncertainty in expected returns dominates uncertainty in the covariance matrix. This is why **robust optimization methods** (shrinkage, Black-Litterman, risk parity) work better than naive mean-variance optimization.

- **Vol mismatch:** If you hedge at implied vol but realized vol is different, you earn the vol spread: $P\&L \propto (\sigma_{\text{realized}}^2 - \sigma_{\text{implied}}^2) \times \Gamma \times S^2$.

Follow-up: “What’s the P&L of a delta-hedged option if realized vol exceeds implied vol?”
 → “You profit. The gamma P&L exceeds theta cost. Specifically, daily P&L $\approx \frac{1}{2} \Gamma S^2 (\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2) \Delta t$.”

Interview: Trading

Q: What is vega and how do volatility traders use it?

A:

Vega measures sensitivity to implied volatility: $\mathcal{V} = \partial V / \partial \sigma = S \phi(d_1) \sqrt{T}$.

How vol traders operate:

1. **View:** “I think implied vol at 25% is too high; realized vol will be 20%.”
2. **Trade:** Sell options (short vega). Delta-hedge with stock.
3. **Risk:** If realized vol is actually 30%, they lose on gamma + vega.

Positions are expressed in “vega-dollars”: “I’m short \$1M vega in 6-month SPX puts.” This means a 1% drop in implied vol earns \$1M.

Important nuance: Vega isn’t constant across strikes. OTM puts have “sticky-delta” vega: as spot drops, the put becomes more ATM, increasing its vega. This creates **vanna** exposure (cross-sensitivity of delta to vol).

Follow-up: “How do you trade the vol smile?” → “Risk reversals (buy put, sell call at different strikes), butterflies (trade curvature), or calendar spreads (trade term structure).”